

Generic Pan Tilt: Open Source Motion Control Platform for Entertainment and Research

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Abstract

We introduce the Generic Pan Tilt, an open-source, two-axis motion control platform designed for use in entertainment, art, and research. Combining affordable off-the-shelf hardware, 3D-printed parts, and custom electronics, the system enables rapid development and flexible integration of kinetic movement into small-scale performances and installations. The Generic Pan Tilt adheres to industry standards for connectivity and control, supporting DMX512-A and modular payloads. Demonstrated in a live AI-augmented musical performance, the platform allows for a new music and performance interfaces that feature expressive motion.

CCS Concepts

• **Hardware** → **Electro-mechanical devices**; Sensor devices and platforms; *Sensors and actuators*; • **Applied computing** → **Performing arts**; Media arts.

Keywords

Electro-mechanical, Hardware, Hardware Platform, Actuator, Kinetic Sculpture, Sculpture, Open Source, Entertainment, Art, Live, Performance, Media

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1 Introduction

The production of live performances and new media installation art often involves components such as sound reinforcement, lighting, graphics, and moving parts. While many of these systems can be found at all sizes and scales of concerts and exhibitions, kinetic sculpture and moving parts remain difficult to implement in smaller productions. Stadium-touring concerts and Broadway musical theater shows often fabricate custom kinetic pieces for each show [1], dramatically raising the minimum implementation cost and expertise. In contrast to these motion systems, existing sound, lighting,

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Figure 1: Generic Pan Tilt fixture with leaf payload

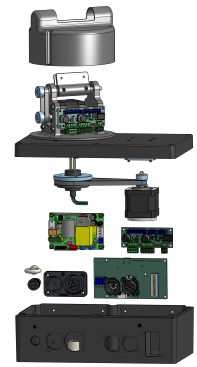


Figure 2: Diagram of lower assembly

and media/graphics systems are available for purchase at varying price points. This may be due in part to the very broad range of types of motion (e.g., winches, tracks, lifts, etc.) that cannot be covered by one product or system [4]. For a smaller production staff or artist studio, implementing a custom system (consisting of power, control, and mechanical components) for each exhibition at lower quantities may still be unfeasible. However, steps may be taken to lower the barrier to entry by releasing an open platform of reusable components and fully-featured implementations.

We present the Generic Pan Tilt, a multi-purpose 2-axis motion fixture. The device provides an open, simple, and inexpensive platform for researchers, artists, and makers to produce actuator systems. The Generic Pan Tilt is similar to moving-head lighting fixtures found in theaters, live concerts, and television studios, but at a fraction of the cost and targets payload flexibility instead of only lighting. It is powered, controlled, and mounted according to the standards of the entertainment industry. Users can choose to reuse the entire electro-mechanical system, or they may implement their own mechanics while applying the electronic parts without any required modifications. The Generic Pan Tilt aims to make kinetic sculpture and motion components more accessible for live performances and exhibitions and enable new interfaces for motion-control applications.

2 Design

The Generic Pan Tilt is designed for rapid assembly and prototyping of entertainment-style motion fixtures at low cost and on a small scale. The design files are open source and published online.¹ It

¹<https://generic-pan-tilt-uist-2025.media.mit.edu/>

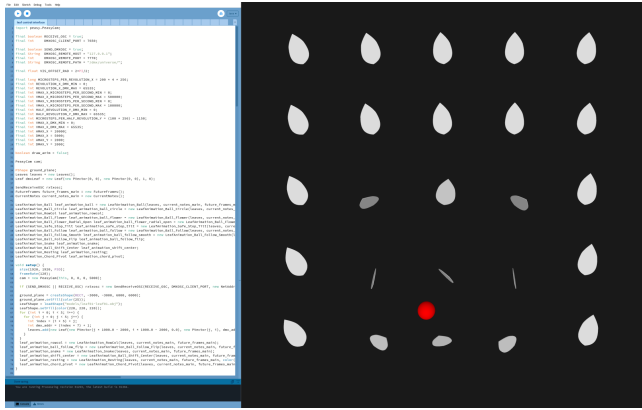


Figure 3: Rendered leaves in Processing application following virtual ball visualizing the AI lead synthesizer

is built from standard off-the-shelf parts, 3D-printed assemblies, and printed circuit boards (PCBs). A diagram is shown in Figure 2. The standard mechanical parts found in the bill of materials can be procured at various qualities and price points to allow for trade-offs between budget and longevity. The 3D-printed assemblies allow for easy iteration and inexpensive production. The PCBs implement commonly found connectivity, reliable motion control, and room for expansion. The total cost per unit in a production of 25 units is about \$200 USD (in 2024).

The main PCB consists of a Raspberry Pi RP2040 microcontroller unit (MCU), isolated RS-485 interface for DMX512-A communication [3], DIP switches for configuration, bicolor status light, debug and USB connections, accelerometer to determine device orientation, and expansion interface. The main board communicates with the motor control and expansion boards through Serial Peripheral Interface (SPI) and i2c (Inter-Integrated Circuit) buses.

Each of the motor control PCBs (one for each axis) contain an Analog Devices TMC5240 stepper motor controller, which handles full position and acceleration control, end stop sensors, encoder inputs, homing, and motor protection. The integrated dynamics of this controller simplifies firmware development by receiving position, velocity, and acceleration commands via the SPI bus instead of precisely timed step and direction pulses from the main MCU.

The fixture implements an infinite rotation pan axis, which is achieved by passing the tilt axis communication buses through a slip ring. The pan axis also features a 3D-printed timing belt pulley: a very inexpensive part compared to an aluminum custom-machined pulley with a large borehole to account for the slip ring. While the pan axis homes magnetically using a hall-effect sensor, the tilt axis homes using feedback from the stepper motor. The tilt axis allows for easy removal of the payload for storage, transport, and reuse.

3 Interface Application

Due to their entertainment-style connectivity, mounting, and controls, the Generic Pan Tilt works well on live performance stages. A system of 25 Generic Pan Tilt fixtures were successfully tested in a performance environment in September 2024 where they accompanied GRAMMY-winning keyboardist and technologist Jordan



Jordan and the JAM_BOT: A Work-in-Progress Performance, 2024. Credit: Caroline Alden. Courtesy of MIT.

Figure 4: Musician Jordan Rudess performing with the Generic Pan Tilt fixtures driven by the AI generated leads

Rudess on stage (Figure 4).² The fixtures were controlled from a Processing [5] application that rendered the motion patterns into the pan and tilt parameters (Figure 3). The leaf-like payload and a range of sudden to subtle movements replicate the motions of leaves blowing in the wind. For this performance, the fixtures physically manifested an AI musical agent that responded to and played along with Rudess in real time. This two-way music and motion interface was a key element of the performance. Working as a choreographed system, the fixtures bridged the gap between the AI and audience, providing insight into the AI performer present onstage. The fixtures provided an expressive medium required to effectively visualize the AI’s musical movements [2].

4 Demonstration

A computer and small piano keyboard running demo programs will allow attendees to interact with live Generic Pan Tilt fixtures in real-time. These demo applications will include:

- a music-based interaction where the AI visualizes a lead synthesizer as it plays along with human musical keyboard input (Figures 3 and 4)
- a mouse-following program
- several pre-programmed animations/patterns
- individual control of the fixture’s input parameters

Our demonstration will consist of a table with two functioning Generic Pan Tilt fixtures mounted above it. These fixtures will be controlled from a MIDI keyboard, MIDI controller, and a mouse. A laptop running the control software will show the rendering interface. A third fixture will also be on the table in an open format to allow inspection of its inner workings. Additional parts (PCBs and 3D-printed parts) will also be displayed. A standing monitor/additional computer will play a video from the September 2024 concert showcasing the emergent patterns and animations possible with the complete fleet of fixtures.

²<https://news.mit.edu/2024/model-virtuosity-jordan-rudess-jam-bot-1119>

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